

Chemical Bonding



- **Why do atoms form bonds?**
- The most stable arrangement of electrons in an atom is to have either a filled valence shell or to have 8 valence electrons (octet rule).
- Atoms obtain this stable electron arrangement by sharing, gaining or losing electrons – which is what creates chemical bonds.

Chemical Bonding



Types of Chemical Bonds

There are three major types of chemical bonds:

- ionic bonding
- covalent bonding
- metallic bonding

Factors that guide the formation of bonds



The type of bond formed depends on the ionization energy and electronegativity of the atoms involved

- **Ionization energy (IE)** = energy required to remove one valence electron.
- metals have LOW ionization energies
- non-metals have HIGH ionization energies
- **Electronegativity (EN)** = measure of the tendency for an atom to “steal” an electron from another atom within a chemical bond. Follows the same trends as electron affinity:
- metals have LOW electronegativities
- non-metals have HIGH electronegativities

The Ionic Bond



- When an atom with a high EN is placed near an atom with a low IE, the first atom may “steal” an electron from the second atom.
- The atom that gives up the electron becomes a positively charged **cation**, and the atom that gains the electron becomes a negatively charged **anion**.
- **An ionic bond** is the “electrostatic” attraction formed between the oppositely charged cation and anion.

Electron transfer



- This electron transfer can only occur if the first atom's electronegativity is quite high, (that is, it readily steals electrons) and the second atom's ionization energy is very low (that is, it readily gives up an electron)!
- Thus, this electron transfer typically occurs between a **metal** (low IE) and a **non-metal** (high EN)

Arrangement of ions in ionic compounds



- Ionic compounds do not form individual “molecules,” composed of just two or three atoms.
- Instead, ionic compounds are composed of literally billions of cations and anions electrostatically attracted to each other.
- The empirical formula for an ionic compound is just the simplest RATIO of the cations to the anions in compound.
- The regular, repeating arrangement of ions in an ionic compound is called the **crystal lattice**.

Properties of Ionic Compounds Ionic Bonding



- Definite crystalline structure
- High melting points
- Brittle as solids
- Most dissolve in water
- Conduct electricity *only* when in the molten state or when dissolved in water.

Properties of Ionic Compounds Ionic Bonding



- **Definite Crystalline Structure:**

This is explained by the factors already discussed concerning the crystal lattice structure: charge of ions, size of ions, and attraction/repulsion of ions.

- **High melting points:**

To melt a substance requires that the particles of the substance be separated, which requires energy to accomplish.

In ionic compounds, each particle is attracted to multiple other particles of opposite charge in a three-dimensional lattice network.

Melting ionic compounds requires breaking **multiple ion-ion attractions, which requires more energy = higher melting pt.**

- **Brittle as Solids:**

If you “hammer” on an ionic crystal, it shatters. Any attempt to distort the crystal lattice causes the “layers” to shift, which causes like-charged ions to become aligned. The resulting repulsive forces shatters the crystal.

Properties of Ionic Compounds Ionic Bonding



- **Dissolve in Water:**

As we shall see later in this chapter, water is a polar molecule. Polar molecules are molecules which have a small degree of charge separation, so that one end of the molecule is slightly negatively charged, and the other end is slightly positively charged.

The partial positive charge on the hydrogen atoms in water are attracted to the anions in an ionic compound; the partial negative charge on the oxygen atom in water are attracted to the cations in an ionic compound.

The water molecule is able to literally “pluck apart” the ionic compound, and carry the ions off into the solution.

Properties of Ionic Compounds Ionic Bonding



- **Conduct Electricity when Molten or Dissolved in Water**

An electric current is made up of electrons in motion. In order for a current to flow through a substance, there must be an electric field, produced when charged particles are separated. (The electric field is what “pushes” the electrons along.)

When an ionic compound is in the solid phase, there is no separation of charges – and no current can flow.

When the compound is dissolved in water or in the molten (liquid) phase, the ions are separated, which creates the electric field and allows the current to flow.

Covalent Bonding



- If two atoms both have relatively high, but similar electronegativities, they can obtain a stable octet of electrons by *sharing electrons between them, rather than transferring electrons from one atom to another*.
- This sharing of electrons by overlapping orbitals between two atoms is called a **covalent bond**.
- Covalent bonds typically form between two non-metals, or between a metalloid and a non-metal.

Molecule



- **A molecule is the name given to two or more atoms which are bonded together by covalent or polar covalent bonds. The atoms may be different elements, or they can all be the same element.**

Polar Covalent Bonding



- A pure covalent bond is an equal sharing of electrons, then we can say a **polar covalent bond is an un-equal sharing of electrons.**
- If the two atoms have nearly identical EN, then the bond is pure covalent. If there is a *large difference in EN between the two bonded atoms, then an ionic bond forms.*
- When there is a moderate difference in EN between the two bonded atoms, a polar covalent bond forms.

Polar molecule



- When the atoms are bonded by polar covalent bonds, the shared electrons are shifted towards the atom with the greater electronegativity. This electron shift creates charge separation within the molecule.
- This charge separation can create a **polar molecule** *in which one end of the molecule has a partial positive charge (δ^+) and the other end has a partial negative charge (δ^-).*

Polar molecule



A — B

- If B is more EN than A, the electrons are shifted away from A and towards B. We represent this shift with an arrow, pointing towards the partial negative side of the molecule, with a cross-bar at the partial positive end.
- Not all molecules with polar covalent bonds are polar molecules. In some cases, the electrons are pulled in opposing directions, which “cancels out” the shift, so that no charge separation occurs:

Properties of Molecular Compounds



- Because molecules can exist as individual, isolated units, many of the properties of the compound often depend on the nature of the attractive forces that exist *between two or more molecules. Note carefully that these attractive forces are NOT covalent bonds.*
- We will investigate the nature of these so-called “intermolecular” attractive forces in more detail later on...

Properties of Molecular Compounds



- Molecular compounds have lower melting points than ionic compounds.
- Molecular compounds are poor conductors of electricity in any phase.
- Molecular compounds have a wide range of solubility in water – some dissolve easily in water, some do not dissolve at all.

Properties of Molecular Compounds



- **Low Melting Points**

The bonding *within a molecule* is a covalent bond between two atoms – however, there are only very weak attractions between separate molecules. As a result, it does not take a lot of energy to separate two molecules (melting).

- **Poor Conductors of Electricity**

Recall that, in order to conduct a current of electricity, you must have charged particles that are some distance apart to create an electric field.

- Molecules are made up of neutral atoms. By sharing the electrons, neither atom has gained or lost electrons to become ions, even when the molecules are separated.
- Thus, there are no charged particles with which to create an electric field, and so they are non-conductors.

Properties of Molecular Compounds

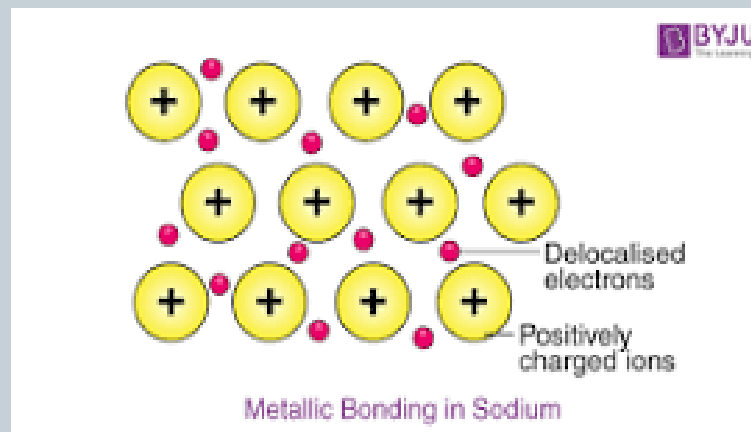


Solubility in Water

- Recall that water is a polar molecule, in which there is a charge separation within the molecule so that one end is partially positive and other end is partially negative.
- Compounds that are also polar molecules can form attractions to water molecules, and they dissolve, much like ionic compounds. Non-polar molecules cannot form attractions to water, and as a result, they do not dissolve in water.

Metallic Bonding

- A metallic bond can be thought of as a mobile “sea” of *delocalized electrons*, which are *mutually attracted to and “shared” by a cluster of metal cations*. The (-) electrons act as the “glue” that holds the(+) cations together.



Properties of metals



- Metals are good conductors of heat & electricity
- Metals are malleable and ductile
- Metals are lustrous

Metals are Good Conductors



- Substances to be a conductor of electricity requires that there be charges that are separated in order to create an electric field. There are (+) cations and (-)electrons that are separated in metallic bonds, which creates the necessary electric field. In addition, the mobile “sea” of electrons themselves are able to move easily = current of electricity.
- Heat is a measure of the transfer of kinetic energy between particles. The mobile e- are able to collide with other e- and cations, transferring kinetic energy as they do so. Thus, metals conduct heat very well.

Metals are Malleable and Ductile



- Malleability and ductility requires that the bonding between atoms be very “flexible” so that, even when the substance is distorted, the atoms are still bonded.
- When a force is applied, the cations are free to move – as long as there are some e⁻ between the cations, the atoms are still bonded together...

Metals are Lustrous



- An object that appears lustrous or shiny is reflecting or emitting multiple photons of light with very similar, but slightly different, energies.
- The delocalized e^- in metals move between shells that are close together in energy levels. This e^- shift produces photons of similar energy = “lustrous” quality of the metal.